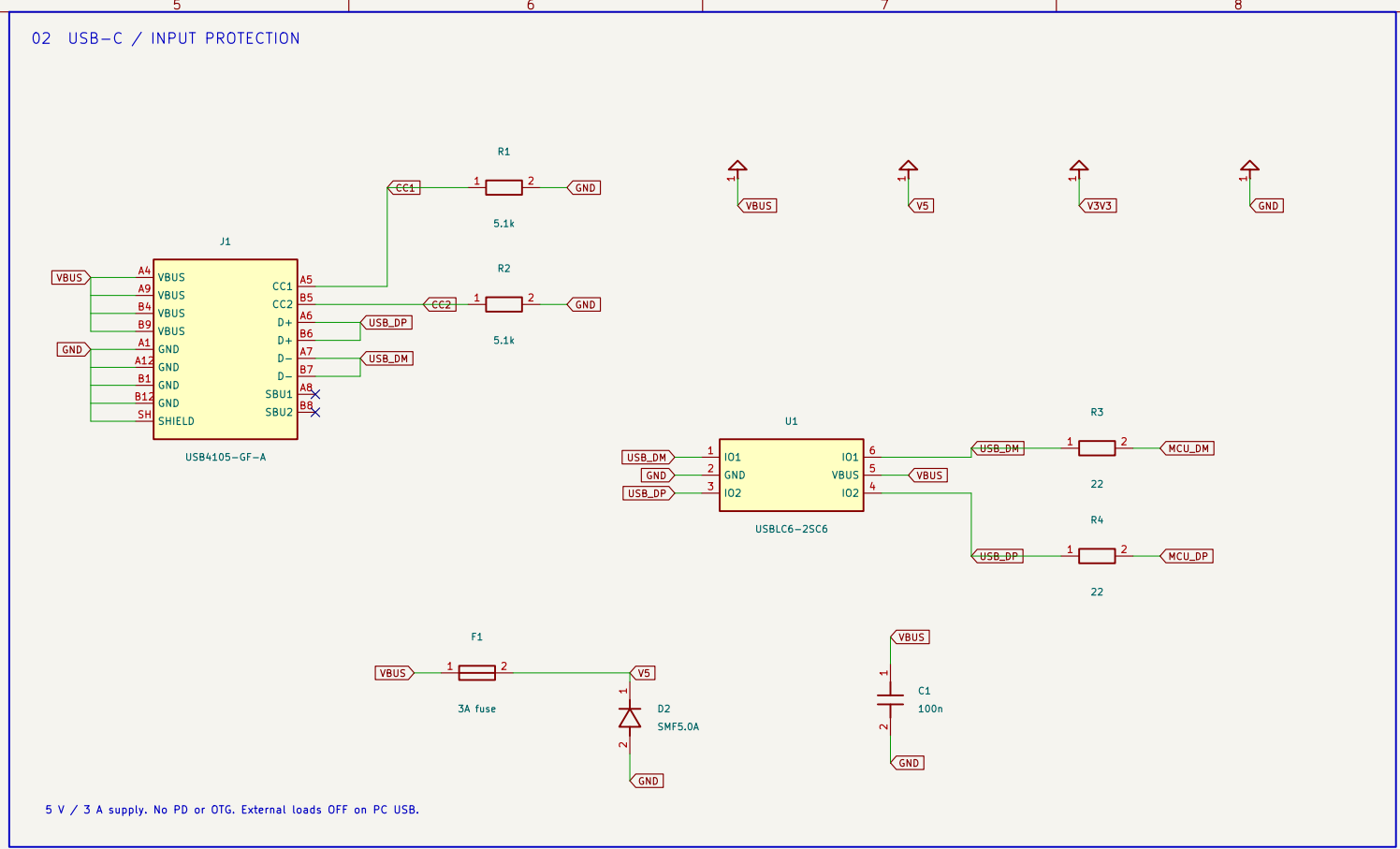


The schematic diagram illustrates the boot circuit for the ESP8266 module. It includes the following components and connections:

- Power Supply:** A V3V3 supply is connected to the EN pin through a 10k resistor (R5). A 100nF capacitor (C7) is connected between V3V3 and GND. A 10uF capacitor (C6) is connected between EN and GND.
- Boot Pin:** The BOOT pin is connected to GND through a 10k resistor (R6). A switch (SW1) is connected between BOOT and GND.
- Module Pinout:** The ESP8266 module (U3) is shown with its pinout. The pins are labeled as follows:
 - 1: IO2
 - 2: GPIO0
 - 3: EN
 - 4: IO3
 - 5: FAULT5
 - 6: IO4
 - 7: IO5
 - 8: V3V3
 - 9: IO7
 - 10: PWR5_EN
 - 11: PWR5_EN
 - 12: IO0
 - 13: GPIO13
 - 14: GPIO14
 - 15: GND
 - 16: IO6
 - 17: FAULT6
 - 18: BOOT
 - 19: GND
 - 20: GPIO26
 - 21: BUZZ_PWM
 - 22: UART_TX
 - 23: GND
 - 24: EPAD
 - 25: GPIO10
 - 26: GPIO28
 - 27: GPIO27
 - 28: GPIO26
 - 29: GPIO25
 - 30: GPIO24
 - 31: GPIO23
 - 32: GPIO22
 - 33: MCU_DM
 - 34: MCU_DP
 - 35: GND
 - 36: IO1
 - 37: FAULT1
 - 38: GND
 - 39: IO2
 - 40: GPIO0
 - 41: EN
 - 42: IO3
 - 43: FAULT3
 - 44: IO4
 - 45: IO5
 - 46: V3V3
 - 47: IO7
 - 48: PWR5_EN
 - 49: PWR5_EN
 - 50: IO0
 - 51: GPIO13
 - 52: GPIO14
 - 53: GND
 - 54: IO6
 - 55: FAULT6
 - 56: BOOT
 - 57: GND
 - 58: GPIO26
 - 59: BUZZ_PWM
 - 60: UART_TX
 - 61: GND
 - 62: EPAD
 - 63: GPIO10
 - 64: GPIO28
 - 65: GPIO27
 - 66: GPIO26
 - 67: GPIO25
 - 68: GPIO24
 - 69: GPIO23
 - 70: GPIO22
 - 71: MCU_DM
 - 72: MCU_DP
 - 73: GND
 - 74: IO1
 - 75: FAULT1
 - 76: GND
 - 77: IO2
 - 78: GPIO0
 - 79: EN
 - 80: IO3
 - 81: FAULT3
 - 82: IO4
 - 83: IO5
 - 84: V3V3
 - 85: IO7
 - 86: PWR5_EN
 - 87: PWR5_EN
 - 88: IO0
 - 89: GPIO13
 - 90: GPIO14
 - 91: GND
 - 92: IO6
 - 93: FAULT6
 - 94: BOOT
 - 95: GND
 - 96: GPIO26
 - 97: BUZZ_PWM
 - 98: UART_TX
 - 99: GND
 - 100: EPAD
 - 101: GPIO10
 - 102: GPIO28
 - 103: GPIO27
 - 104: GPIO26
 - 105: GPIO25
 - 106: GPIO24
 - 107: GPIO23
 - 108: GPIO22
 - 109: MCU_DM
 - 110: MCU_DP
 - 111: GND
 - 112: IO1
 - 113: FAULT1
 - 114: GND
 - 115: IO2
 - 116: GPIO0
 - 117: EN
 - 118: IO3
 - 119: FAULT3
 - 120: IO4
 - 121: IO5
 - 122: V3V3
 - 123: IO7
 - 124: PWR5_EN
 - 125: PWR5_EN
 - 126: IO0
 - 127: GPIO13
 - 128: GPIO14
 - 129: GND
 - 130: IO6
 - 131: FAULT6
 - 132: BOOT
 - 133: GND
 - 134: GPIO26
 - 135: BUZZ_PWM
 - 136: UART_TX
 - 137: GND
 - 138: EPAD
 - 139: GPIO10
 - 140: GPIO28
 - 141: GPIO27
 - 142: GPIO26
 - 143: GPIO25
 - 144: GPIO24
 - 145: GPIO23
 - 146: GPIO22
 - 147: MCU_DM
 - 148: MCU_DP
 - 149: GND
 - 150: IO1
 - 151: FAULT1
 - 152: GND
 - 153: IO2
 - 154: GPIO0
 - 155: EN
 - 156: IO3
 - 157: FAULT3
 - 158: IO4
 - 159: IO5
 - 160: V3V3
 - 161: IO7
 - 162: PWR5_EN
 - 163: PWR5_EN
 - 164: IO0
 - 165: GPIO13
 - 166: GPIO14
 - 167: GND
 - 168: IO6
 - 169: FAULT6
 - 170: BOOT
 - 171: GND
 - 172: GPIO26
 - 173: BUZZ_PWM
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 - 175: GND
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 - 179: GPIO27
 - 180: GPIO26
 - 181: GPIO25
 - 182: GPIO24
 - 183: GPIO23
 - 184: GPIO22
 - 185: MCU_DM
 - 186: MCU_DP
 - 187: GND
 - 188: IO1
 - 189: FAULT1
 - 190: GND
 - 191: IO2
 - 192: GPIO0
 - 193: EN
 - 194: IO3
 - 195: FAULT3
 - 196: IO4
 - 197: IO5
 - 198: V3V3
 - 199: IO7
 - 200: PWR5_EN
 - 201: PWR5_EN
 - 202: IO0
 - 203: GPIO13
 - 204: GPIO14
 - 205: GND
 - 206: IO6
 - 207: FAULT6
 - 208: BOOT
 - 209: GND
 - 210: GPIO26
 - 211: BUZZ_PWM
 - 212: UART_TX
 - 213: GND
 - 214: EPAD
 - 215: GPIO10
 - 216: GPIO28
 - 217: GPIO27
 - 218: GPIO26
 - 219: GPIO25
 - 220: GPIO24
 - 221: GPIO23
 - 222: GPIO22
 - 223: MCU_DM
 - 224: MCU_DP
 - 225: GND
 - 226: IO1
 - 227: FAULT1
 - 228: GND
 - 229: IO2
 - 230: GPIO0
 - 231: EN
 - 232: IO3
 - 233: FAULT3
 - 234: IO4
 - 235: IO5
 - 236: V3V3
 - 237: IO7
 - 238: PWR5_EN
 - 239: PWR5_EN
 - 240: IO0
 - 241: GPIO13
 - 242: GPIO14
 - 243: GND
 - 244: IO6
 - 245: FAULT6
 - 246: BOOT
 - 247: GND
 - 248: GPIO26
 - 249: BUZZ_PWM
 - 250: UART_TX
 - 251: GND
 - 252: EPAD
 - 253: GPIO10
 - 254: GPIO28
 - 255: GPIO27
 - 256: GPIO26
 - 257: GPIO25
 - 258: GPIO24
 - 259: GPIO23
 - 260: GPIO22
 - 261: MCU_DM
 - 262: MCU_DP
 - 263: GND
 - 264: IO1
 - 265: FAULT1
 - 266: GND
 - 267: IO2
 - 268: GPIO0
 - 269: EN
 - 270: IO3
 - 271: FAULT3
 - 272: IO4
 - 273: IO5
 - 274: V3V3
 - 275: IO7
 - 276: PWR5_EN
 - 277: PWR5_EN
 - 278: IO0
 - 279: GPIO13
 - 280: GPIO14
 - 281: GND
 - 282: IO6
 - 283: FAULT6
 - 284: BOOT
 - 285: GND
 - 286: GPIO26
 - 287: BUZZ_PWM
 - 288: UART_TX
 - 289: GND
 - 290: EPAD
 - 291: GPIO10
 - 292: GPIO28
 - 293: GPIO27
 - 294: GPIO26
 - 295: GPIO25
 - 296: GPIO24
 - 297: GPIO23
 - 298: GPIO22
 - 299: MCU_DM
 - 300: MCU_DP
 - 301: GND
 - 302: IO1
 - 303: FAULT1
 - 304: GND
 - 305: IO2
 - 306: GPIO0
 - 307: EN
 - 308: IO3
 - 309: FAULT3
 - 310: IO4
 - 311: IO5
 - 312: V3V3
 - 313: IO7
 - 314: PWR5_EN
 - 315: PWR5_EN
 - 316: IO0
 - 317: GPIO13
 - 318: GPIO14
 - 319: GND



07 HEADER ESD / SHUNT PROTECTION

03 SGM6232 / ALWAYS-ON 3.3 V

EN open: auto-start. EP to GND. FB divider $33k / 10.5k = 3.314 V$.

The left diagram shows the internal pull-up configuration for comparator U4. The IN pin is connected to PWR5_EN through a resistor R9 (10k) and a capacitor C9 (100nF). The EN pin is connected to GND. The GND pin is connected to GND. The OUT pin is connected to V3_SW through a capacitor C11 (1uF). The ILIM pin is connected to GND through a resistor R7 (26.1k). The FAULT_N pin is connected to GND.

The right diagram shows the internal pull-up configuration for comparator U5. The IN pin is connected to PWR5_EN through a resistor R10 (10k) and a capacitor C10 (100nF). The EN pin is connected to GND. The GND pin is connected to GND. The OUT pin is connected to V5_SW through a capacitor C12 (1uF). The ILIM pin is connected to GND through a resistor R8 (26.1k). The FAULT_N pin is connected to GND.

Fault inputs: firmware must enable internal pullups.

TP5

FAULT_N 1

TP6

FAULT_N 1

26.1k ILIM: nominal 0.989 A; bounds 0.908–1.081 A.

05 PASSIVE 5020 BUZZER

The schematic shows a buzzer circuit. A green trace from VDD3 passes through a diode D3 (SS34) and connects to pin 1 of buzzer BZ1 (KLJ-5020 3.3V). Pin 2 of BZ1 is connected to GND. The buzzer is also driven by a PWM signal BUZZ_PWM through resistor R11 (100 ohms) to pin 1 of driver Q1 (A03A004 MOSFET). The gate of Q1 is labeled BUZZ_GATE and its drain is connected to pin 2 of BZ1. The source of Q1 is connected to GND. A capacitor C13 (100nF) is connected between VDD3 and GND.

D3
SS34

BZ1
KLJ-5020 3.3V

R11
100

BUZZ_PWM

BUZZ_GATE

Q1
A03A004

C13
100n

VDD3

GND

4 kHz / 50% duty. GPIO12 reserved.

ix12 2.54mm right-angle

U6/U7: four channels each: no supply or bypass needed.

I00-I07: 3.3 V only: final-board ESD test required.

U10: switched rails. Power current bypasses the array.

Place arrays and bypass capacitors directly beside J2.

Short, wide ground returns: qualify ESD on final PCB.

I00-I07: 3.3 V only; final-board ESD test required.

U10: switched rails. Power current bypasses the array.

Place arrays and bypass capacitors directly beside J2.

Short, wide ground returns; qualify ESD on final PCB.

Place arrays and bypass capacitors directly beside J2.

Short, wide ground returns: qualify ESD on final PCB.

Short, wide ground returns: qualify ESD on final PCB.

Short, wide ground returns; quality ESD on final PCB.